Example 3.33 Solve $2x^2 - 3x - 3 = 0$ by formula method.

Solution Compare $2x^2 - 3x - 3 = 0$ with the standard form $ax^2 + bx + c = 0$

$$a = 2,\ b = -3,\ c = -3$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

substituting the values of a, b and c in the formula we get,

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(2)(-3)}}{2(2)} = \frac{3 \pm \sqrt{33}}{4}$$

Therefore, $x = \frac{3 + \sqrt{33}}{4}, \frac{3 - \sqrt{33}}{4}$

Example 3.34 Solve $3p^2 + 2\sqrt{5}p - 5 = 0$ by formula method.

Solution Compare $3p^2 + 2\sqrt{5}p - 5 = 0$ with the Standard form $ax^2 + bx + c = 0$

$$a = 3,\ b = 2\sqrt{5},\ c = -5\,.$$

$$p = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

substituting the values of a, b and c in the formula we get,

$$p = \frac{-2\sqrt{5} \pm \sqrt{\left(2\sqrt{5}\right)^2 - 4(3)(-5)}}{2(3)} = \frac{-2\sqrt{5} \pm \sqrt{80}}{6} = \frac{-\sqrt{5} \pm 2\sqrt{5}}{3}$$

Therefore, $x = \frac{\sqrt{5}}{3}, -\sqrt{5}$

Example 3.35 Solve $pqx^2 - (p+q)^2 x + (p+q)^2 = 0$

Solution Compare the coefficients of the given equation with the standard form $ax^2 + bx + c = 0$

$$a = pq,\ b = -(p+q)^2,\ c = (p+q)^2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

substituting the values of a, b and c in the formula we get,

$$x = \frac{-\left[-(p+q)^2\right] \pm \sqrt{\left[-(p+q)^2\right]^2 - 4(pq)(p+q)^2}}{2pq}$$

$$= \frac{(p+q)^2 \pm \sqrt{(p+q)^4 - 4(pq)(p+q)^2}}{2pq}$$

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